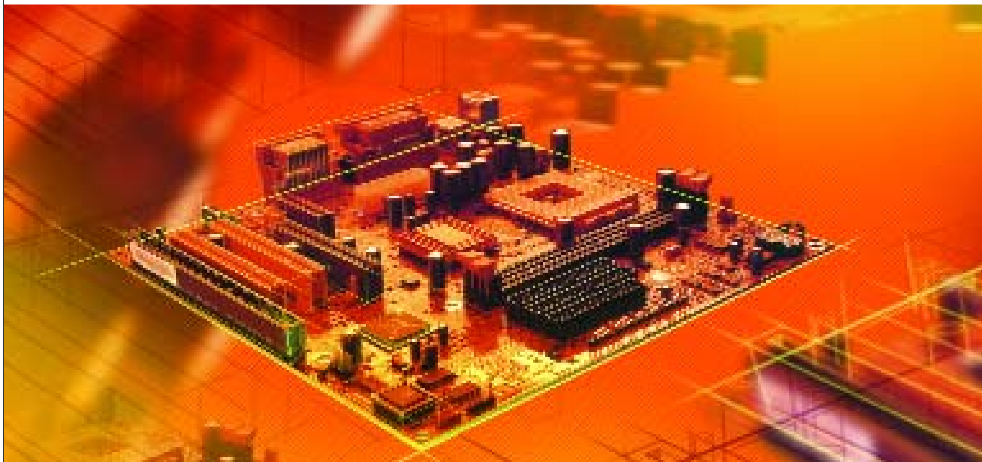

Motherboards

Ready Reckoner

Low-end **You Need**	A motherboard that will support older hardware and has newer features
Look for	nForce2 chipset-based motherboards for AMD's Athlon XP and Intel's 865 chipset for an Intel Processor
Mid-end **You Need**	A motherboard for good performance supporting some of the latest features
Look for	An nForce 3 chipset for Athlon XPs and Intel's 875P chipset forPentium 4 processors
High-end **You Need**	A motherboard that is a great performer and is feature packed
Look for	nForce 4 chipset board for AMD 64-bit and Intel's 925X chipset based-motherboards

The computer enthusiast never had it better, since everything has gotten faster, cheaper, and easier to use. New products will integrate the CPU, graphics, and the entire chipset, often called "System on a Chip (SOC)".

They still can't integrate enough memory yet, so memory will continue to be in external DRAM chip form. So far, these SOC devices are meant for low-end applications such as Web Pads, PDAs, and other smart "Information Appliances".

However, this approach represents the eventual endpoint for PC integration.

Newer technology on the heels of DDR DRAM is a natural evolution to Quad-Rate DRAM, pumping data out 4 times on each clock. Motherboards supporting such RAM chips will be tested and proven on servers before coming to desktops.

Facts

- Investing in a better motherboard provides better future upgrade solutions, as well as performance.
- The maximum configuration to which a PC can be upgraded including the type of processor, amount of RAM, size of hard disk and external add-on cards depends on the motherboard.
- Certain motherboards support overclocking—by increasing the FSB of your processor the speed of the processor increases; exceeding the speed too much may result in CPU failure though.

Facts

- Motherboards that support two processors can perform two functions simultaneously.
- Before you buy motherboard keep upgradeability in mind, especially the graphics subsystem and the processor speed it can support.
- If you are building a new system based on 915 or 925 chipset, lookout for the combo boards that support both DDR and DDR2 standard memories.
- Motherboards based on ATi's chipset for AMD's 64 bit processor are supposed to hit the market by this December, so wait and watch. These motherboards promise much better onboard graphics than what we see on today's motherboards.
- If you are buying feature loaded enthusiast's motherboard make sure you buy a really good power supply that has enough juice to supply the demand.
- Overclocker's should pay special attention to the quality of MOSFET's and capacitors used on the board. Better the quality better the overclocking potential of the board.
- When buying dual processor motherboard pay attention to the placement of the two processor sockets; there should be considerable space between them for proper ventilation.

Buying Tips

- Building a new PC or overhauling an old one? The motherboard is the place to start. Your motherboard determines the type of RAM and CPU you can use; look for future-proof motherboards that provide extra features such as FireWire (IEEE 1394) and over four USB 2.0 ports, six-channel sound chips, and either SATA or IDE RAID controllers.
- Buying a Motherboard supporting CPUs that are not going to be phased out soon makes sense—during your next upgrade you won't need to change your motherboard. Motherboards supporting DDR and DDR2 memory are an added advantage, as DDR is cost effective—and DDR2 gives better performance.
- When it comes to top performers and the latest technology, look for nForce 4 chipset boards for AMD 64-bit processors with business-calibre RAID functionality, 802.11g Wireless PCI card, IEEE 1394b, an additional SATA RAID controller, 7.1-channel audio, a PCI Express x16 slot for video cards and a dual BIOS. For the latest motherboard from Intel, look for Intel's 925X chipset-based boards—these support features such as PCI-Express; DDR2 memory up to 533 MHz; High-Definition Audio subsystem; and SATA support.
- Motherboards based on Intel's 865 and 875 series chipsets are things of the past—invest on boards such as the 925 series for Pentium processors, and nForce 3 or nForce 4 for AMD's 64-bit processors.